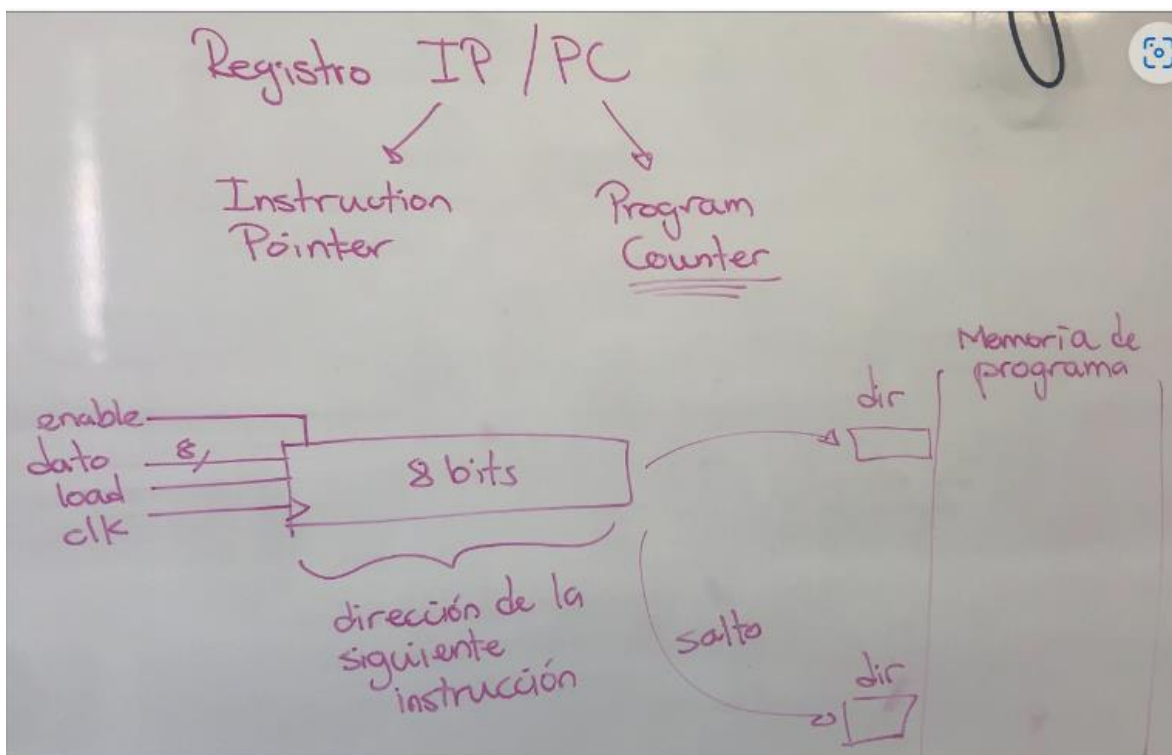


Joshua Alexis Máñez Alcayde
243990

Practice #6: Counters with Verilog in Cadence

Design a program counter or instruction pointer with Verilog in Cadence, based on the next diagram.



The circuit functionality is represented in the next table:

enable	load	clk	Q (8bits)
1	0	↓	$Q + 1$
x	1	↓	dato
otro caso			$Q_{anterior}$

Joshua Alexis Máynez Alcayde

243990

HDL code

```
1 //Verilog HDL for "Verilog", "Counter" "functional"
2
3
4 module Counter (Enable,Data,Load,Clk,Reset,qout);
5
6 input Enable;
7 input [7:0] Data;
8 input Load;
9 input Clk;
10 input Reset;
11 output [7:0] qout;
12 reg [7:0] q;
13 reg [7:0] Out_s;
14
15 always @(posedge Clk) begin
16     if(Load == 1'b0 && Enable == 1'b1)begin
17         q <= q+1;
18     end
19     else if (Load == 1'b1)begin
20         q <= Data;
21     end
22     else if(Reset)begin
23         q <= 8'b0;
24     end
25 end
26
27 assign qout = q;
28
29 endmodule
30
```

Test bench codes

```
1 //Verilog HDL for "Verilog", "Counter_tb" "functional"
2
3
4 module Counter_tb ();
5
6 reg Enable;
7 reg [7:0] Data;
8 reg Load;
9 reg Clk;
10 reg Reset;
11 wire [7:0] qout;
12
13 Counter uut(Enable,Data,Load,Clk,Reset,qout);
14
15 initial begin
16
17     $display("Start");
18
19     Clk = 1;
20     Enable = 0;
21     Reset = 0;
22     Load = 1;
23     Data = 4'b0000; //D
24     #10;
25     Clk = 0;
26
27
28     #10;
29     Clk = 1;
30
31     #10;
32     Clk = 0;
33
34     Reset = 1;
35     Clk = 0;
36     Data = 4'b1101; //D
37     Enable = 1;
38     Load = 0;
39     Reset = 0;
40     #10;
```

```
41   Clk = 1;
42   #10;
43   Clk = 0;
44   #10;
45   Clk = 1;
46
47   #10;
48   Clk = 0;
49   Data = 4'b0011; //3
50   Enable = 1;
51   Load = 0;
52   Reset = 0;
53   #10;
54   Clk = 1;
55   #10;
56   Clk = 0;
57   #10;
58   Clk = 1;
59
60   #10;
61   Clk = 0;
62   Enable = 0;
63   Load = 1;
64   Reset = 0;
65   #10;
66   Clk = 1;
67   #10;
68   Clk = 0;
69   #10;
70   Clk = 1;
71
72   #10;
73   Clk = 0;
74   Enable = 1;
75   Load = 0;
76   Reset = 0;
77   #10;
```

```
78   Clk = 1;
79   #10;
80   Clk = 0;
81   #10;
82   Clk = 1;
83
84   #10;
85   Clk = 0;
86   #10;
87   Clk = 1;
88   #10;
89   Clk = 0;
90   #10;
91   Clk = 1;
92   #10;
93   Clk = 0;
94   #10;
95   Clk = 1;
96   #10;
97   Clk = 0;
98   #10;
99   Clk = 1;
100
101   Enable = 0;
102   Load = 0;
103
104   #10;
105   Clk = 0;
106   Data = 4'b1111; //F
107   Enable = 1;
108   Load = 0;
109   Reset = 0;
110   #10;
111   Clk = 1;
112   #10;
113   Clk = 0;
```

Joshua Alexis Máynez Alcayde
243990

```
114     #10;
115     Clk = 1;
116     Enable = 0;
117     Load = 0;
118     #10;
119     Clk = 0;
120     Enable = 0;
121     Load = 0;
122     Reset = 1;
123     #10;
124     Clk = 1;
125     #10;
126     Clk = 0;
127     #10;
128     Clk = 1;
129
130     #10;
131     Clk = 0;
132     #10;
133     Clk = 1;
134     #10;
135     Clk = 0;
136     #10;
137     Clk = 1;
138     $display("Test completed");
139
140 end
141
142 endmodule
143
```

Simulation



Joshua Alexis Máynez Alcayde
243990

CODES

```
module Counter (Enable,Data,Load,Clk,Reset,qout);
```

```
input Enable;  
input [7:0] Data;  
input Load;  
input Clk;  
input Reset;  
output [7:0] qout;  
reg [7:0] q;  
reg [7:0] Out_s;
```

```
always @(posedge Clk) begin  
    if(Load == 1'b0 && Enable == 1'b1)begin  
        q <= q+1;  
    end  
    else if (Load == 1'b1)begin  
        q <= Data;  
    end  
    else if(Reset)begin  
        q <= 8'b0;  
    end  
end
```

```
assign qout = q;
```

```
endmodule
```

```
/////////////////////////////////////////////////////////////////  
module Counter_tb ();
```

```
reg Enable;  
reg [7:0] Data;  
reg Load;  
reg Clk;  
reg Reset;  
wire [7:0] qout;
```

```
Counter uut(Enable,Data,Load,Clk,Reset,qout);
```

```
initial begin
```

```
    Clk = 1;  
    Enable = 0;
```

Joshua Alexis Máynez Alcayde
243990

```
Reset = 0;  
Load = 1;  
Data = 4'b0000; //D  
#10;  
Clk = 0;
```

```
#10;  
Clk = 1;
```

```
#10;  
Clk = 0;
```

```
Reset = 1;  
Clk = 0;  
Data = 4'b1101; //D  
Enable = 1;  
Load = 0;  
Reset = 0;  
#10;  
Clk = 1;  
#10;  
Clk = 0;  
#10;  
Clk = 1;
```

```
#10;  
Clk = 0;  
Data = 4'b0011; //3  
Enable = 1;  
Load = 0;  
Reset = 0;  
#10;  
Clk = 1;  
#10;  
Clk = 0;  
#10;  
Clk = 1;
```

```
#10;  
Clk = 0;  
Enable = 0;  
Load = 1;  
Reset = 0;
```


Joshua Alexis Máynez Alcayde
243990

```
#10;  
Clk = 1;  
#10;  
Clk = 0;  
#10;  
Clk = 1;
```

```
#10;  
Clk = 0;  
Enable = 1;  
Load = 0;  
Reset = 0;  
#10;  
Clk = 1;  
#10;  
Clk = 0;  
#10;  
Clk = 1;
```

```
#10;  
Clk = 0;  
#10;  
Clk = 1;  
#10;  
Clk = 0;  
#10;  
Clk = 1;  
#10;  
Clk = 0;  
#10;  
Clk = 1;  
#10;  
Clk = 0;  
#10;  
Clk = 1;
```

```
Enable = 0;  
Load = 0;
```

```
#10;  
Clk = 0;  
Data = 4'b1111; //F  
Enable = 1;  
Load = 0;
```

Joshua Alexis Máynez Alcayde
243990

```
    Reset = 0;
    #10;
    Clk = 1;
    #10;
    Clk = 0;
    #10;
    Clk = 1;
    Enable = 0;
    Load = 0;
    #10;
    Clk = 0;
    Enable = 0;
    Load = 0;
    Reset = 1;
    #10;
    Clk = 1;
    #10;
    Clk = 0;
    #10;
    Clk = 1;

    #10;
    Clk = 0;
    #10;
    Clk = 1;
    #10;
    Clk = 0;
    #10;
    Clk = 1;
    $display("Test completed");

end

endmodule
```